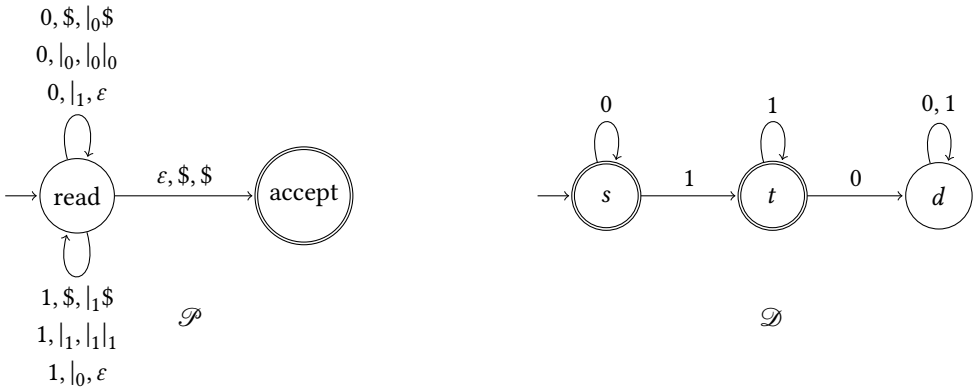
CS 2490 – Homework 9

Turn in by 2026/11/14 at 11:59pm

You **must** turn in your assignment as a zip file entitled: *lastname_firstname_hw9.zip*. It will contain a pdf file *lastname_firstname_hw9.pdf* with your solutions to the exercises, and a JSON file *lastname_firstname_hw9_ex1.json* for the PDA of Exercise 1.

Exercise 1. In this exercise, we work over the alphabet $\Sigma = \{0, 1\}$. Recall the language $E = \{w \in \Sigma^* \mid n_0(w) = n_1(w)\}$ of Session 24, and the PDA[1] $\mathscr{P}$ of Session 26 that recognizes it. Here is $\mathscr{P}$, together with a DFA $\mathscr{D}$ that recognizes $\mathrm{L}(0^*1^*)$:

[1] The stack alphabet is $\Gamma = \{\$, |_0, |_1\}$ and the initial stack symbol is \$. The symbol $|_0$ remembers an unmatched 0, and $|_1$ an unmatched 1.



(i) What is the language $E \cap \mathrm{L}(0^*1^*)$? Which theorem of Session 29 guarantees that it is context-free?

(ii) Draw the product PDA $\mathscr{P} \times \mathscr{D}$ of Session 29.

Include the JSON file of this PDA (produced with the web application) in your submission.

(iii) Which states of $\mathscr{P} \times \mathscr{D}$ are useless? Justify your answer.

Exercise 2. In this exercise, we work over the alphabet $\Sigma = \{a, b, c, d\}$. Consider the two languages

$$L_1 = \{a^n b^k c^n d^k \mid n, k \geq 0\}, \qquad L_2 = \{a^n b^k c^k d^n \mid n, k \geq 0\}.$$

(i) Show that L_1 is not context-free.

(ii) Show that L_2 is context-free. In other words, exhibit a context-free grammar that generates L_2 and give a quick justification.

(iii) Since L_2 is context-free, the argument you used in (i) can't work for L_2. To see where it fails, take $w = a^p b^p c^p d^p \in L_2$, and exhibit a factorization $w = xuyvz$, with $|uyv| \leq p$ and $|uv| \geq 1$, such that $xu^i yv^i z \in L_2$ for every $i \geq 0$.[2]

[2] Recall from Session 34 that the pumping length given by the proof of the lemma is $p = r^{m+1} \geq 2^2 = 4$.

(iv) Using (i) and a closure property of context-free languages, show that the language

$$M = \{w \in \Sigma^* \mid n_a(w) = n_c(w) \text{ and } n_b(w) = n_d(w)\}$$

is not context-free.[3]

[3] *Hint: look at the second proof of Example 6 in Session 34.*

Exercise 3. In this exercise, we work over the alphabet $\Sigma = \{a, b, c\}$. Show that the language

$$L_3 = \{w \in \Sigma^* \mid n_a(w) \times n_b(w) = n_c(w)\}$$

is not context-free. For example, *cab*, *babcc* and *aaa* are in L_3, but *ab* and *abcc* are not.

Exercise 4. In this exercise, we work over the alphabet $\Sigma = \{a, b, c\}$. Show that the language

$$L_4 = \{a^i b^j c^k \mid i, j, k \geq 0 \text{ and } i^2 + j^2 = k^2\}$$

is not context-free.[4] For example, $a^3b^4c^5$ and b^2c^2 are in L_4, but *abc* is not.

[4] *Hint: try $w = a^{3p}b^{4p}c^{5p}$. But be careful: for some factorizations, pumping 0 times stays in L_4, and for others, pumping 2 times does.*